

Food-intake detection from AIM-2 sensor signals: literature review, model description and study design

Caelan Dunlea
University of Alabama

30 September 2026

1 Literature review

Objective measurement of eating behaviour matters for obesity, eating disorders and diabetes research, and self-report is known to be inaccurate and cannot capture in-meal behaviour such as eating rate or pauses (1). Wearable sensors that detect chewing, bites or hand gestures have therefore become a central approach, ranging from classical machine learning on hand-crafted features to end-to-end deep learning, with the aims of improving accuracy, reducing false detections and generalising to people outside the training set (2). Fontana et al. (3) introduced the Automatic Ingestion Monitor (AIM), combining jaw-motion, hand-gesture and accelerometer sensors, and reported 89.8% food-intake detection accuracy over 24 hours of free living. Its successor, AIM-2 (4), moved the sensors onto eyeglasses: a temporalis-muscle sensor and a 3-axis accelerometer at 128 Hz. It extracted 38 hand-crafted features per signal, selected nine by minimum-redundancy maximum-relevance, and classified 10 s epochs with a support vector machine, reaching F1 0.818 under leave-one-subject-out validation on 30 participants, as summarised in Table 1. These feature-engineered models work, but their performance is bounded by the features chosen by hand, and they are sensitive to noise and to variation between people.

Table 1: State-of-the-art methods for food-intake and chewing detection, and the two ECG studies whose methods transfer to this problem.

References	Techniques	Goals	Findings
Doulah et al. (4)	AIM-2; hand-crafted features, mRMR selection, SVM	Food-intake detection with passive food-image capture	F1 0.818 (LOSO, 30 participants); camera triggered only during eating
Papapanagiotou et al. (5)	1D CNN on raw in-ear audio	Chewing detection without hand-crafted features	Accuracy 0.980, F1 0.883
Kyritsis et al. (6)	CNN + LSTM, end to end on raw wrist IMU	Bite and meal detection in the wild	Bite F1 0.923 (LOSO)

References	Techniques	Goals	Findings
Ghosh and Sazonov (7)	Five deep networks (MLP, time-CNN, FCN, ResNet, Inception) on AIM-2 optical and accelerometer time series	Compare deep architectures for food-intake detection	ResNet best, balanced accuracy 93.47%
Ghosh et al. (8)	Wavelet scalogram of accelerometer, fused with food-image recognition	Reduce false positives in free living	Fused F1 80.77%; fusion cuts false positives
Stankoski et al. (9)	Selection of hard negative examples for training	Imbalanced data with imperfect labels	Person-independent F1 0.82
Wang et al. (10)	Temporal CNN + multi-head self-attention	Eating speed from wrist IMU, towards free living	61 participants, 513 h; eating-speed MAPE 0.110
Vedovelli et al. (11)	Transformer encoder and other models on smartwatch sensors	Eating-episode recognition	Transformer: sensitivity 0.691, specificity 0.504 (LOSO)
Hannun et al. (12)	34-layer 1D CNN on raw single-lead ECG	End-to-end arrhythmia classification	F1 0.837, above the average cardiologist (0.780)
Ikram et al. (13)	Transformer on MIT-BIH beats	Five-class arrhythmia classification	Accuracy 97.1%; split not at patient level

1.1 From hand-crafted features to learned representations

To address the limits of hand-crafted features, several groups trained networks directly on the raw signal. Papapanagiotou et al. (5) detected chewing with a 1D convolutional network applied to raw in-ear audio windows, and Kyriasis et al. (6) trained a CNN feature extractor and an LSTM jointly on raw wrist inertial data, reaching a bite-detection F1 of 0.923 under leave-one-subject-out evaluation. On AIM-2 itself, Ghosh and Sazonov (7) compared five deep architectures on the optical and accelerometer time series and found the residual network best, with a balanced accuracy of 93.47%. The same pattern holds outside eating: the fully convolutional network of Wang et al. (14), three blocks of convolution, batch normalisation and ReLU, is a strong baseline across time-series benchmarks.

The closest analogue to this problem is electrocardiography. Hannun et al. (12) classified arrhythmia with a 34-layer 1D CNN applied end to end to raw single-lead ECG at 200 Hz, and exceeded the average cardiologist (F1 0.837 against 0.780). ECG and the AIM-2 signals share a sampling rate of the same order, a quasi-periodic structure (heart beats at about 1 Hz, chews at 0.94–2.17 Hz (15)) and the need to recognise short events inside longer recordings. This motivates the first and most important design choice in this work: the model reads the time series, not a spectrum. Ghosh et al. (8) took the opposite route, converting accelerometer signals to a wavelet scalogram image. A magnitude spectrum says which rhythms are present in a window but discards when they occur, which is the information that separates a bite from background motion; a scalogram keeps a time axis, but at a resolution fixed by the transform rather than learned from the data.

1.2 Attention and Transformers

Attention mechanisms let a network weight the parts of its input that carry evidence. Squeeze-and-excitation (16) gates feature channels with a small learned function of their global average, at negligible cost. Attention-based pooling (17) treats a window as a bag of instances and learns which instances determine its label; for eating, the instances are time steps and a single bite may be the deciding one. Transformers extend this to attention between all time steps. Ikram et al. (13) applied a Transformer to five-class arrhythmia classification on MIT-BIH, reporting 97.1% accuracy. The reported 80/20 split is of preprocessed samples rather than of patients, so the figure measures performance on new beats from known patients, not on new patients. In eating detection, Wang et al. (10) combined a temporal CNN with multi-head self-attention to measure eating speed from wrist IMU data on 61 participants, while Vedovelli et al. (11) found that a Transformer encoder on smartwatch data under leave-one-subject-out gave the highest sensitivity (0.691) but a specificity of only 0.504, which is the false-positive problem in its clearest form.

Despite their promise, Transformers carry many more parameters than the attention layers above, and whether they help at a given data size is an empirical question rather than a given. With 20 annotated sessions (7.8 h), this work uses the two light attention mechanisms and treats a Transformer over the convolutional features as an experiment to run once the annotated data has grown (Section 3).

1.3 Supervision, augmentation and negative data

Small annotated datasets call for more supervision per example and for more examples per recording. Multi-task learning (18) improves generalisation by training a shared representation on related tasks; here, counting chews is the related task, since the ground truth already contains chew annotations. Augmentation of wearable sensor data raised accuracy from 77.54% to 86.88% in Parkinson’s disease monitoring (19); overlapping windows are the simplest form of it. Window length itself trades resolution for context, and has a measurable effect on recognition (20).

False positives are the persistent weakness of eating detectors: many non-eating activities produce jaw and head motion. Stankoski et al. (9) reduced them by deliberately selecting hard negative examples for training, and Ghosh et al. (8) by fusing the sensor classifier with food-image recognition. Finally, evaluation practice varies widely across the field (2). Saeb et al. (21) showed that record-wise cross-validation, where windows from one person appear in both training and test sets, can massively overestimate accuracy, and that subject-wise validation is the one that matches the use case. The leave-one-subject-out protocol used by Doulah et al. (4) and Diou et al. (1) is adopted here for the same reason.

1.4 Contribution of this work

- A compact convolutional model with channel attention, temporal attention pooling and an auxiliary chew-count head, reading raw AIM-2 optical and accelerometer signals directly from the ETS database, evaluated leave-one-subject-out.
- A controlled ablation showing that the input representation accounts for most of the performance: replacing the time series with an FFT magnitude spectrum costs 0.087 F1 and 0.180 precision, while the attention and auxiliary components are within noise at the current data size.
- Two documented negative results with a mechanism: temporal median smoothing and un-marked

context windows both lower F1, because meals at 8 s resolution are punctuated rather than contiguous.

- A measured check of sensor-annotation alignment on every session, replacing an assumed correction with a verified one.

Despite the progress in deep learning for intake monitoring, several challenges persist: limited annotated data, variation between participants in sensor fit and chewing style, false positives from non-eating jaw and head motion, and inconsistent evaluation that makes published results hard to compare. Addressing these requires models that generalise to people they have not seen, evaluated in the way they would be used.

2 Model description

The model is a compact one-dimensional convolutional network that reads the raw AIM-2 sensor signals and outputs the probability that an 8 s window contains eating (Figure 1). It has four parts. A convolutional feature extractor learns its own filters from the time series instead of receiving hand-crafted features or a spectrum. A squeeze-and-excitation gate reweights the learned feature channels in each window. An attention pooling layer weights the time steps that carry the evidence. Finally, the output has two heads: one classifies eating, and one counts chews and is used only during training. The network has 47,282 parameters at inference (Table 2). Table 3 lists every design decision with its reason and the literature it rests on.

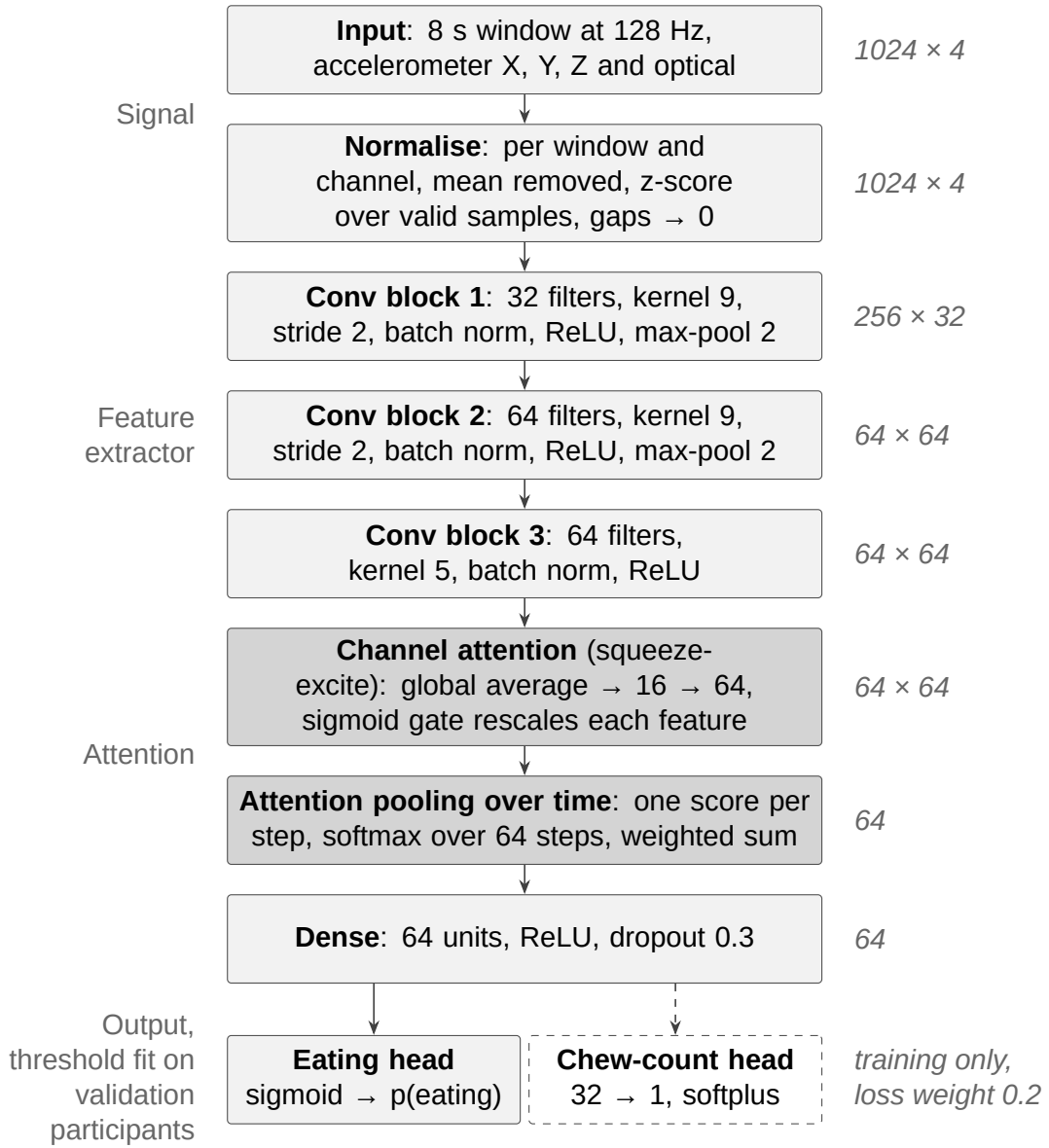


Figure 1: Model architecture. Shaded blocks are the two attention mechanisms. The dashed chew-count head is trained jointly and discarded at inference. Output shapes are time steps $\times$ features.

Table 2: Layer dimensions. An 8 s window at 128 Hz is 1024 samples; each of the 64 output time steps covers about 0.9 s of signal at 125 ms spacing.

Layer	Configuration	Output	Parameters
Input	Acc X, Y, Z, Optical	1024 × 4	–
Conv block 1	32 × k9, stride 2, BN, max-pool 2	256 × 32	1,312
Conv block 2	64 × k9, stride 2, BN, max-pool 2	64 × 64	18,752
Conv block 3	64 × k5, BN	64 × 64	20,800
Channel attention	64 → 16 → 64, sigmoid	64 × 64	2,128
Attention pooling	1 × 1 conv, softmax over time	64	65
Dense	64, ReLU, dropout 0.3	64	4,160
Eating head	1, sigmoid	1	65
<i>Total at inference</i>			47,282
Chew-count head	32 → 1, softplus (training only)	1	2,113

Table 3: Complete description of the food-intake detection model. Every design decision, the reason for it, and the literature it is grounded in.

#	Component	Design	Why	Literature
<i>Data</i>				
1	Sensor	AIM-2 eyeglass sensor: 3-axis accelerometer and optical temporalis-muscle sensor, 128 Hz, stored as 8 s packets in the ETS database	Chewing moves the temporalis muscle and the head; both channels carry it, with different artefacts. The original AIM-2 used a flex sensor where this one has the optical sensor	Doulah et al. (4), Ghosh and Sazonov (7)
2	Labels	Eating = bout OR bite annotation at 10 Hz. Window is eating if more than half its annotated samples are eating. Unrecorded time (−1) is dropped, never treated as non-eating	Treating unannotated time as negative would bury the real labels and invent easy negatives	(this work)
3	Alignment	No time shift. Sensor/annotation lag measured by cross-correlating the optical chewing envelope with the labels: median +0.10 s, all 20 sessions within 0.2 s	A correction must be measured, not assumed; a residual lag of one packet would corrupt every label edge	(this work)

continued on next page

#	Component	Design	Why	Literature
4	Window	8 s (1024 samples), non-overlapping at evaluation	7 to 17 chew cycles at typical chewing rates (0.94–2.17 Hz); matches the packet size. Window length has a measurable effect and is swept in Section 3	Po et al. (15), Banos et al. (20)
<i>Input</i>				
5	Representation	Raw time series, 1024 samples $\times$ 4 channels, no hand-crafted features and no spectrum	A magnitude spectrum keeps which rhythms are present but discards when they occur; a bite is a transient and chewing an envelope. Largest single effect in the ablation (F1 –0.087 with FFT input)	Ghosh and Sazonov (7), Hannun et al. (12), Wang et al. (14)
6	Normalisation	Per window, per channel: mean removed, z-scored over valid samples only; missing samples set to 0 after scaling	Optical baseline varies by participant and fit on the face; DC level is not information about chewing. A literal –1 against counts near 3000 injects a step at every gap	Ismail Fawaz et al. (22)
<i>Network</i>				
7	Convolutional trunk	Three 1D conv blocks: 32@k9/s2, 64@k9/s2, 64@k5, each with batch norm and ReLU, max-pool 2 after the first two. Output 64 time steps $\times$ 64 features	Learns its own band-pass filters from data. Each output step covers about 0.9 s of signal (one chew cycle) at 125 ms spacing	Ghosh and Sazonov (7), Wang et al. (14)
8	Channel attention	Squeeze-and-excitation: global average $\rightarrow$ dense 16 (ReLU) $\rightarrow$ dense 64 (sigmoid) $\rightarrow$ rescale each feature channel	Optical carries chewing; accelerometers carry motion that is sometimes signal and often artefact. Reweights per window at negligible cost	Hu et al. (16)
9	Temporal pooling	Attention pooling: one score per time step (1 $\times$ 1 conv), softmax over the 64 steps, weighted sum	Global averaging dilutes a 1–2 s bite across 8 s. Attention weights the informative steps, as in multiple-instance learning	Ilse et al. (17)
10	Classifier	Dense 64 (ReLU), dropout 0.3, sigmoid output	Small head; about 47 k parameters in total at inference	–

continued on next page

#	Component	Design	Why	Literature
11	Auxiliary head	Chew-count regression (dense 32 $\rightarrow$ 1, softplus), target scaled by its 99th percentile. Used in training only	One label bit per window is thin supervision; counting chews asks the shared trunk to represent the chewing rhythm	Caruana (18)
<i>Training</i>				
12	Loss	Binary cross-entropy (weight 1.0) + chew MSE (weight 0.2); class-balanced sample weights computed per fold	Auxiliary loss is a regulariser, not the objective. Eating fraction ranges 17–74% by session, so the rare class changes with the fold	Caruana (18)
13	Augmentation	50% overlapping windows (4 s hop) at training time only	Roughly doubles the training windows at no collection cost; evaluation stays on the non-overlapping grid so no second is scored twice	Um et al. (19)
14	Optimisation	Adam, learning rate 0.001, batch 64, up to 80 epochs, early stopping (patience 12) and learning-rate halving on validation loss	Standard; validation participants are held out from training, never random windows	–
<i>Evaluation</i>				
15	Validation	Leave-one-subject-out: each participant is the test set once. Inner validation on two further held-out participants	Windows of one meal are near-duplicates; any split below participant level leaks and overstates performance	Diou et al. (1), Saeb et al. (21)
16	Threshold	Chosen on the inner validation participants at the centre of the F1 plateau; never on the test participant	The F1 curve is flat near its peak, so the exact argmax on two participants is noise	–
17	Post-processing	None. Median smoothing over 3 windows was tested and rejected (F1 -0.079)	Meals are punctuated by pauses; at 8 s the label genuinely switches, and smoothing erases short real bouts	(this work)
18	Metrics	F1 and balanced accuracy lead; accuracy, precision, recall reported alongside, pooled over all held-out windows	Class balance differs by participant; accuracy alone rewards predicting the majority class	Bell et al. (2)

2.1 Current performance

Under leave-one-subject-out validation on the 20 annotated sessions available at the time of writing, pooled over all 3,503 held-out non-overlapping windows, the model reaches F1 0.838, accuracy 0.890, balanced accuracy 0.879, precision 0.828 and recall 0.848. Per-participant F1 varies considerably (mean 0.811, standard deviation 0.124), which is the expected cost of leave-one-subject-out and the honest measure of how the model generalises to a new person. These figures will be regenerated on the 61 sessions now annotated.

3 Proposed study design

The literature in Section 1 points to one gap more than any other: food-intake detectors from wearable chewing sensors are usually reported as a single architecture with a single number, and the design decisions behind them are rarely isolated. The ablation in Table 4 already shows that those decisions are not equally important, since the input representation accounts for most of the performance and the attention components cannot be separated from noise. The proposed study makes that the question rather than a side result: which design dimensions matter for deep food-intake detection from a chewing sensor, measured one at a time under leave-one-subject-out validation.

Table 4: Preliminary ablation: the model with one component removed or added. 20 annotated sessions, participant-level 4-fold cross-validation, same folds for every row. Fold-to-fold spread in F1 is about 0.048, so effects smaller than that are not distinguishable from noise. To be repeated under leave-one-subject-out on the enlarged dataset.

Variant	F1	Change	Precision	Verdict
Full model	0.854	–	0.876	–
FFT magnitude input instead of time series	0.767	–0.087	0.696	Hurts
No overlapping training windows	0.829	–0.025	0.834	Possibly hurts
No attention (global average pooling)	0.847	–0.007	0.857	Within noise
No chew-count head	0.849	–0.006	0.857	Within noise
Add median smoothing (3 windows)	0.775	–0.079	0.765	Hurts
Add 8 s context each side, no marker	0.775	–0.080	0.683	Hurts

3.1 Protocol, fixed for every experiment

- **Leave-one-subject-out validation** on all annotated participants, following Saeb et al. (21). With 20 to 61 sessions this is computationally feasible, and it is the closest approximation of the use case: a participant the model has never seen.
- **Paired comparison per participant.** Each variant is scored on the same held-out participants as the reference model, so the difference can be tested per participant (Wilcoxon signed-rank) rather than read against the spread of fold means.
- **Three seeds per configuration**, reported as mean and range, so initialisation noise is not mistaken for an effect.

- **False positives per hour of non-eating** reported alongside F1, because that is the failure mode that matters in free-living use and the one negative data (below) is meant to fix.

3.2 Dimensions to explore

Table 5: Planned experiments. Each changes one dimension of the model in Table 3 and holds everything else fixed.

Dimension	Levels	Question	Grounding
Published architectures	FCN, ResNet and Inception as compared by Ghosh and Sazonov (7), trained on the same folds	Where does this model stand against the closest prior work on AIM-2, under an identical protocol?	Ghosh and Sazonov (7), Wang et al. (14)
Input representation	Time series, FFT magnitude, wavelet scalogram	Does keeping temporal structure explain the gain, and does a time-frequency image recover it?	Ghosh et al. (8), Hannun et al. (12)
Sampling rate	128, 64, 32, 16 Hz	How much of the signal is needed? Chewing lies below about 2.2 Hz, so 32 Hz still leaves a Nyquist margin of seven times.	Po et al. (15)
Window length	2, 4, 8, 16 s	Resolution against context: shorter windows localise bites, longer windows contain more chew cycles.	Banos et al. (20)
Context	None; 8 s each side with a target-window marker channel	Can surrounding signal help without the model answering "is there eating anywhere nearby?"	(this work)
Chew-head weight	0, 0.1, 0.2, 0.5, 1.0	How strongly should counting chews shape the features, and does it help participants with weak optical signal?	Caruana (18)
Temporal model	Attention pooling; small Transformer encoder over the 64 convolutional time steps	Does self-attention between time steps add to a convolutional front end at this data size?	Ikram et al. (13), Zerveas et al. (23), Zeng et al. (24)
Self-supervised pre-training	None; convolutional trunk pre-trained on unannotated AIM-2 recordings in ETS	Can the unannotated sensor data, most of what the database holds, stand in for labels the study does not have?	Diou et al. (1)

Negative data	Annotated sessions only; plus non-eating recordings from other ETS studies	Do additional negative participants reduce false positives without costing recall?	Bell et al. (2), Ghosh et al. (8)
---------------	--	--	-----------------------------------

The published-architecture row comes first: it is the comparison a reviewer will ask for, and it replaces comparisons with the lab's earlier model. The representation, sampling-rate and window rows are cheap and answer the questions raised at the 9/23 lab meeting directly. The Transformer row is the one most likely to be asked about by reviewers, and the one most at risk of overfitting on a dataset this size; it should be run only after the dataset has been enlarged to 61 sessions, and reported whether it helps or not. Negative data is expected to have the largest practical effect, since false positives are the model's main weakness, and is the natural joint experiment with the lab.

References

- [1] Christos Diou, Konstantinos Kyritsis, Vasileios Papapanagiotou, and Ioannis Sarafis. Intake monitoring in free-living conditions: Overview and lessons we have learned. *Appetite*, 176:106096, 2022. doi: 10.1016/j.appet.2022.106096.
- [2] Brooke M. Bell, Ridwan Alam, Nabil Alshurafa, Edison Thomaz, Abu S. Mondol, Kayla de la Haye, John A. Stankovic, John Lach, and Donna Spruijt-Metz. Automatic, wearable-based, in-field eating detection approaches for public health research: a scoping review. *npj Digital Medicine*, 3: 38, 2020. doi: 10.1038/s41746-020-0246-2.
- [3] Juan M. Fontana, Muhammad Farooq, and Edward Sazonov. Automatic ingestion monitor: A novel wearable device for monitoring of ingestive behavior. *IEEE Transactions on Biomedical Engineering*, 61(6):1772–1779, 2014. doi: 10.1109/TBME.2014.2306773.
- [4] Abul Doulah, Tonmoy Ghosh, Delwar Hossain, Masudul H. Imtiaz, and Edward Sazonov. “Automatic Ingestion Monitor Version 2” – a novel wearable device for automatic food intake detection and passive capture of food images. *IEEE Journal of Biomedical and Health Informatics*, 25(2): 568–576, 2021. doi: 10.1109/JBHI.2020.2995473.
- [5] Vasileios Papapanagiotou, Christos Diou, and Anastasios Delopoulos. Chewing detection from an in-ear microphone using convolutional neural networks. In *39th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, pages 1258–1261, 2017. doi: 10.1109/EMBC.2017.8037060.
- [6] Konstantinos Kyritsis, Christos Diou, and Anastasios Delopoulos. A data driven end-to-end approach for in-the-wild monitoring of eating behavior using smartwatches. *IEEE Journal of Biomedical and Health Informatics*, 25(1):22–34, 2021.
- [7] Tonmoy Ghosh and Edward Sazonov. A comparative study of deep learning algorithms for detecting food intake. In *44th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, pages 2993–2996, 2022. doi: 10.1109/EMBC48229.2022.9871278.
- [8] Tonmoy Ghosh, Yue Han, Viprav Raju, Delwar Hossain, Megan A. McCrory, Janine Higgins, Carol Boushey, Edward J. Delp, and Edward Sazonov. Integrated image and sensor-based food intake detection in free-living. *Scientific Reports*, 14:1665, 2024. doi: 10.1038/s41598-024-51687-3.

- [9] Simon Stankoski, Marko Jordan, Hristijan Gjoreski, and Mitja Luštrek. Smartwatch-based eating detection: Data selection for machine learning from imbalanced data with imperfect labels. *Sensors*, 21(5):1902, 2021. doi: 10.3390/s21051902.
- [10] Chunzhuo Wang, T. Sunil Kumar, Walter De Raedt, Guido Camps, Hans Hallez, and Bart Vanrumste. Eating speed measurement using wrist-worn IMU sensors towards free-living environments. *IEEE Journal of Biomedical and Health Informatics*, 28(10):5816–5828, 2024.
- [11] Luca Vedovelli, M. J. Bhuyan, Corrado Lanera, Ileana Baldi, and Dario Gregori. Recognition of eating episodes via commercial smartwatch sensors. *PLOS Digital Health*, 2026. doi: 10.1371/journal.pdig.0001539.
- [12] Awni Y. Hannun, Pranav Rajpurkar, Masoumeh Haghpanahi, Geoffrey H. Tison, Codie Bourn, Mintu P. Turakhia, and Andrew Y. Ng. Cardiologist-level arrhythmia detection and classification in ambulatory electrocardiograms using a deep neural network. *Nature Medicine*, 25(1):65–69, 2019. doi: 10.1038/s41591-018-0268-3.
- [13] Sunnia Ikram, Amna Ikram, Harvinder Singh, Malik Daler Ali Awan, Sajid Naveed, Isabel De la Torre Díez, Henry Fabian Gongora, and Thania Candelaria Chio Montero. Transformer-based ECG classification for early detection of cardiac arrhythmias. *Frontiers in Medicine*, 12:1600855, 2025. doi: 10.3389/fmed.2025.1600855.
- [14] Zhiguang Wang, Weizhong Yan, and Tim Oates. Time series classification from scratch with deep neural networks: A strong baseline. In *International Joint Conference on Neural Networks (IJCNN)*, pages 1578–1585, 2017. doi: 10.1109/IJCNN.2017.7966039.
- [15] J. M. C. Po, J. A. Kieser, L. M. Gallo, A. J. Tésenyi, P. Herbison, and M. Farella. Time-frequency analysis of chewing activity in the natural environment. *Journal of Dental Research*, 90(10), 2011. doi: 10.1177/0022034511416669.
- [16] Jie Hu, Li Shen, and Gang Sun. Squeeze-and-excitation networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 7132–7141, 2018. doi: 10.1109/CVPR.2018.00745.
- [17] Maximilian Ilse, Jakub M. Tomczak, and Max Welling. Attention-based deep multiple instance learning. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, volume 80 of *Proceedings of Machine Learning Research*, pages 2127–2136, 2018.
- [18] Rich Caruana. Multitask learning. *Machine Learning*, 28:41–75, 1997. doi: 10.1023/A:1007379606734.
- [19] Terry T. Um, Franz M. J. Pfister, Daniel Pichler, Satoshi Endo, Muriel Lang, Sandra Hirche, Urban Fietzek, and Dana Kulić. Data augmentation of wearable sensor data for parkinson’s disease monitoring using convolutional neural networks. In *Proceedings of the 19th ACM International Conference on Multimodal Interaction (ICMI)*, pages 216–220, 2017. doi: 10.1145/3136755.3136817.
- [20] Oresti Banos, Juan-Manuel Galvez, Miguel Damas, Hector Pomares, and Ignacio Rojas. Window size impact in human activity recognition. *Sensors*, 14(4):6474–6499, 2014. doi: 10.3390/s140406474.
- [21] Sohrab Saeb, Luca Lonini, Arun Jayaraman, David C. Mohr, and Konrad P. Kording. The need to approximate the use-case in clinical machine learning. *GigaScience*, 6(5), 2017. doi: 10.1093/gigascience/gix019.

- [22] Hassan Ismail Fawaz, Germain Forestier, Jonathan Weber, Lhassane Idoumghar, and Pierre-Alain Muller. Deep learning for time series classification: a review. *Data Mining and Knowledge Discovery*, 33(4):917–963, 2019. doi: 10.1007/s10618-019-00619-1.
- [23] George Zerveas, Srideepika Jayaraman, Dhaval Patel, Anuradha Bhamidipaty, and Carsten Eickhoff. A transformer-based framework for multivariate time series representation learning. In *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pages 2114–2124, 2021. doi: 10.1145/3447548.3467401.
- [24] Ailing Zeng, Muxi Chen, Lei Zhang, and Qiang Xu. Are transformers effective for time series forecasting? In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pages 11121–11128, 2023. doi: 10.1609/aaai.v37i9.26317.